University of Padua

Department of Mathematics "Tullio Levi-Civita"

Master Degree in Computer Science

GeoMedia: A Social Media for Location-Based and time-sensitive content sharing

Master's Thesis

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Summary

Over the past few decades, the use of mobile devices has increased drastically, almost replacing desktop software and web browsers. This shift has been mainly driven by the widespread adoption of smartphones not just for phone capabilities but also for the availability of integrated sensors such as GPS, gyroscopes, and Bluetooth, which allow richer and more context-aware user experiences. By leveraging these technologies, Online Social Network (OSN) platforms have been able to provide more advanced and sophisticated features; consequently collecting larger amounts of metadata, interactions and daily engagement.

This thesis presents "GeoMedia", an open-source framework designed for sharing multimedia content — such as text, images, audio, and files — anchored to real-world geographic locations. By leveraging GPS data available on mobile devices, the application provides an interactive map through which users can explore and publish digital content, creating a seamless integration between the digital and physical worlds. Moreover, the framework offers several features and covers various use cases, including restricting content visibility within a customizable distance range, enabling time-sensitive content or applying user-based restrictions for content creation and visualization. It also supports sequential visibility between published posts thus satisfying various usage scenarios and discovery features among published data.

Furthermore, it analyzes the motivations behind GeoMedia, its critical aspects and design considerations, describes its system architecture and implementation through a comparison of different technological paradigms. Finally, it explores potential applications and outlines possible future developments of the framework.

A previous version of this framework has been presented at "GoodIT 2025: International Conference on Information Technology for Social Good" (September 3-5, 2025 - Antwerp, Belgium) [1]

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Chapter 1

Introduction

1.1 Overview

Technology has always influenced society, from the first introduction of the Personal Computer to the Internet, changing the world year after year; from personal life to workspaces, people are surrounded by software on their computers and mobile devices. Over the last decades, the investments in the development of technological innovations, both in hardware and software, as increased exponentially, shaping economical and social impact by integrating the latest features into everyday life. Smartphones and tablets have been the core engine that completely changed daily habits and digital usage; their wide adoption has reached almost three quarters of global population [48, 43], making users abandon traditional computers, expecially for social media[42] and instant messaging platform[45], this trend bring a change on the design and development of applications into a 'mobile first' logic.

Compared to computers, mobile devices offer distinct functionalities and strengths that have facilitated their widespread adoption. These can be categorized according to the fundamental logic underlying any technology:

1. **Hardware**: the increase in computational performance[30], battery life[6], fast connectivity, and the integration of several built-in sensors[29] such as gyroscopes, GPS, NFC, light sensors, cameras, and vibration motors. These enable applications to interact with different fields such as health[4], education[35], and, through motion-sensors or cameras, an interaction with physical world, creating a more immersive user experience.

2. **Software**: the development and presence of mobile applications has played a crucial role. In fact, platforms such as 'Windows Phone' were eventually abandoned and discontinued due to the lack of applications [5, 49] compared to operating systems like 'iOS'[12] and 'Android'[15].

In this way, smartphones apps have became a key integration with daily life eand physical world, for example, when planning a trip or just driving, smartphones, integrated with cars assists users via app like "Google Maps"[17] or "Waze"[10], providing not just directions but also real-time traffic updates, read events or information about available parking. This makes the surrounding environment enriched with digital details. Another example is 'Google Lens'[16], which can identify objects from a photograph, or "Shazam"[41] that through devices' microphone can recognize a playing song. Even in workplaces or public areas, technology integrations can be present, from the resturant that provide a digital menu by a tablet, to customers checking the nutrition macros into a grocery store by scanning the barcode on the pack, to museums by scanning a QR Code enabling audio-guided tours or unlock Augmented Reality functionalities.

1.2 Online Social networks

Alongside the widespread adoption of smartphones and mobile devices, online social networks, have emerged as one of the most influential and important sectors of the technology market. Today, social media platforms rank among the firsts most used applications worldwide [36, 11], while the companies that provide them become some of the most bigger actors in global financial markets [37, 26, 22].

Online Social Networks (OSN) are the digitalized form of a Social Network (typically represented as a graph of people where links correspond to relationships) through the Internet[38]. For instance, Facebook, one of the largest platforms, represents a graph of friendships, where nodes (users) are connected via undirected edges, assuming that friendships are mutual. There are also different types of OSN, such as Instagram or Twitter, where the links between users are directed, implementing a "following" model.

These services provide a variety of functionalities, first of all instant messaging capabilities, both one-to-one and one-to-many, and, in addition, the ability to share a wide range of multimedia content, including images, videos, audio recordings, and other types of documents.This facility and advantages compared to other communication forms, like SMS, has led to their integration across diverse social, professional, and public contexts. The particularity of sharing multimedia content, commonly referred to as a 'post', has coined the term Social Media, where the published content represent the node of interaction between users through comments or reactios (e.g., 'like' or 'unlike'), thus to replicate human behavior in dialogues and social interactions.

Despite the significant benefits offered by social media services, including enhanced connectivity, community creation, and information sharing, their adoption has also introduced various social and ethical challenges. Concerns regarding user privacy are particularly relevant, as it is often not respected by the service providers themselves, who collect personal information and interests in order to monetize them or enable surveillance practices[27]. Other serious issues include the dissemination of misinformation and fake news, nowadays further amplified by Artificial Intelligence-generated content, commonly known as 'deepfakes' [28]. Such content can influence public opinion and social behavior, not always in a positive way [47, 31].

Online Social Networks have also intensified common societal problems through phenomena such as 'echo chambers'[3], where users are repeatedly exposed to information and content that aligns with their existing beliefs and preferences. This continuous reinforcement can strengthen existing views, reduce exposure to alternative perspectives, and, in some cases, contribute to ideological radicalization. In practice, examples can be observed in so-called 'incel' communities [32] and among supporters of extreme political movements.

To address these issues, social media platforms have adopted various moderation and safety mechanisms, including automated content analysis systems, community guidelines and reporting tools. While these measures contribute reducing the spread of harmful content, they are not always fully effective, due to the scale and complexity of emerging innovations, such as 'deepfake' [2]. Consequently user reporting and human moderation aside of automated tools, continue to play a crucial role in identifying, evaluating, and removing content that violates policies or legal regulations[8].

Chapter 2

State of Art

GeoMedia positions itself among existing online social media platforms, by offering functionality centered on the sharing of geolocated content linked to specific latitude and longitude coordinates. Content is visible only within a configurable distance radius and can be more restricted with time-sensitive parameters or user-based policy. Thus to create a wide range of real-world applications, including scavenger hunts, interactive city guides, and any other location-based experiences

The platform enables users to discover and interact with the physical environment through a map enriched by content catagorized into different topics collections, contributed by other users, who can upload pictures or other multimedia resources associated with specific locations.

Although several other social media platforms utilize the GPS capabilities of mobile devices to enhance user experience and provide location-aware features, none fully realize the underlying concept of GeoMedia or offer the same degree of flexibility for creating diverse application scenarios and exploration of uploaded content.

2.1 Foursquare Swarm

Foursquare Swarm is the spin-off of a formerly app, "Foursquare City Guide", which provide personalized recommendations of nearby places to the users' current location based on its interests or previous browsing history and check-ins (locations previously visited by the user).

Foursquare was discontinued in 2024 and its functionalities have been integrated into Swarm, which in addition allows users to share their position with relatives, maintaining a personal lifelog of visited places, which can be shared with others.

For each visited location, users may leave reviews or upload multimedia content; however, these contributions are strictly tied to physical place, such as stores, cafés, restaurants or other kind of attractions as shown in Figure 2.1. Users cannot publish content at arbitrary geographic coordinates, nor can restrict post visibility, as all content is publicly accessible. The application further encourages the usage through a gamification system, in which users earn badges for visiting locations and uploading content. These mechanisms are broadly comparable to those implemented in other platforms such as Google Maps (see Section 2.4).

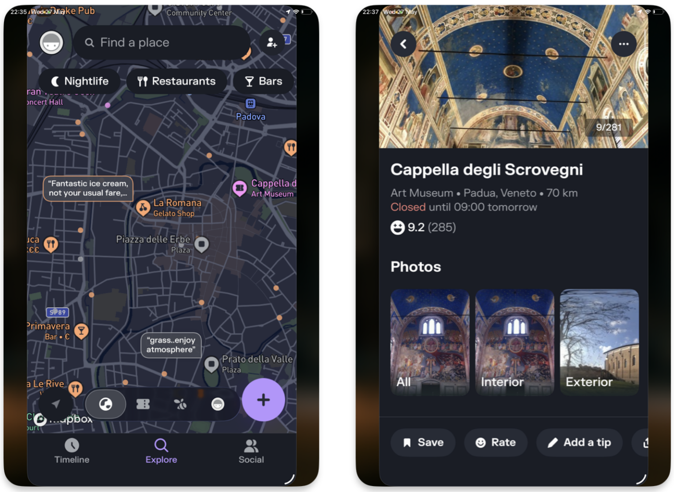

Figure 2.1: Foursquare Swarm show case

2.1.1 Dodgeball and Google Latitude

Dodgeball was a service that simply allowed users to share their current locations with friends through a map, and served as a precursor to Foursquare City Guide and Foursquare Swarm same functionality. Later was acquired by Google and rebranded as 'Google Latitude' [20] discontinued in 2013 and integrated into another discontinued social media by Google (Google+ [18]) and finally migrated into Google Maps 2.4 in 2018.

2.2 Snapchat

Snapchat [25] has been one of the most used social media platforms worldwide [24], mainly thanks to its instant messaging service and multimedia functionality, such as time-based content, which makes uploaded content visible only for 24 hours or visible only once to a user. This encourages users to check it daily in order not to miss anything and to keep up with relatives' lives. As a negative effect, these features have fueled some social impacts such as FoMO (Fear of Missing Out) [9, 40], leading users to become increasingly attached to the social platform and also creating self-comparison with content posted by other users. This problem is not strictly related to Snapchat, but is very common in every self-centric social media platform. Other concerns are raised by privacy and safety dued to the possibility to share current location into a geographic map [33], as done by Swarm and others 2.1, Snapchat also enrich the map by showing geographic closer content to user, grouped into a unified cluster of information, as illustrated in Figure 2.2.

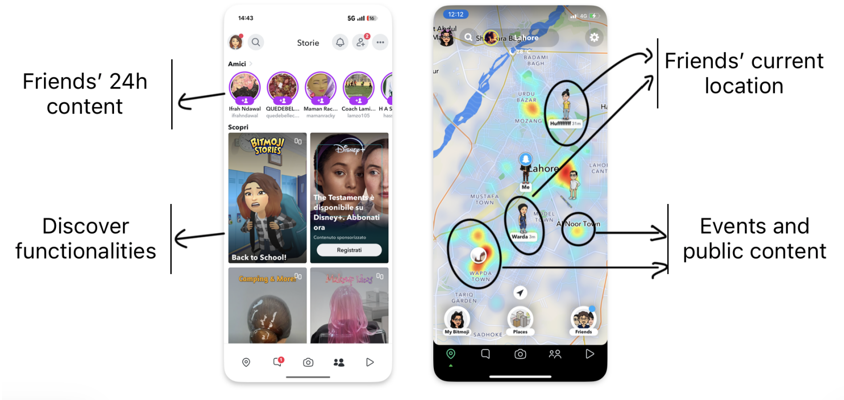

Figure 2.2: Snapchat functionalities

Moreover, as the Swarm app does 2.1, it allows users can log visited locations, including public places such as bars, stores, and museums. This enables the association of user-generated content with specific points of interest (POI) and supports the aggregation of geographically and contextually related content to represent events, such as concerts or other real-word gatherings, as illustrated in Figure 2.3

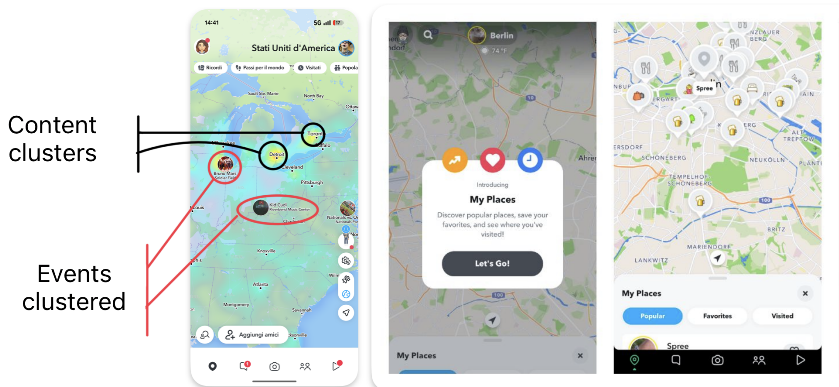

Figure 2.3: Snapchat 'my place' functionality

Snapchat represents a notable example of the adoption of the "mobile-first" paradigm, expecially in social media applications. The platform leverages not only GPS module but also a variety of mobile device sensors, including the camera for augmented reality (AR) experiences and the microphone to records voice-based content. By integrating these hardware features into its core functionality, the app delivers a highly immersive and personalized user experience.

2.3 Instagram

Instagram, owned by Meta Platforms, is one of the biggest and most used social media of the latest decades. Through the introduction of numerous features and continuous innovations, it has attracted more than two billion montly active users worldwide [39, 44].

The platform was initially released in October 2010 exclusively for iOS devices. In fact, at the time, the uploaded content was framed in a square (1:1) aspect ratio with a resolution of 640 pixels, reflecting the display characteristics of iPhones at the time. Later acquired by the formerly named Facebook Inc. (now Meta Platforms Inc.), the application was also deployed for Android systems in April 2012. Then, the original aspect-ratio limitations were removed, and several additional functionalities were introduced, including direct messaging and support for multiple images or videos within a single post.

A major development was the introduction of the "Stories" feature, which provide the same mechanism created by Snapchat 2.2, that allows users to publish content that remains visible for only 24 hours after publication. The addition of this ability contributed significantly to Instagram surpassing Snapchat in user engagement metrics by 2019 [46].

To enrich the media sharing, Instagram provides a variety of creative tools, including photographic filters, image-editing capabilities, and even a live-streaming functionality. The platform also popularized the use of hashtags as a content categorization mechanism, enabling users to discover media and connect with communities centered around shared interests and topics.

Instagram to encourage content discovery adopt a sophisticated recommendation systems that leverage contextual information, including geographic location derived from GPS or cellular network data. Such metadata$_g$ contributes to the generation of personalized recommendations, allowing users to discover nearby profiles, locations, and content that align with their interests [23].

Despite its popularity, Instagram has been the subject of significant criticism. Concerns have been raised regarding user privacy, large-scale behavioral profiling, and the potential influence of recommendation algorithms on user behavior. Additional controversies involve the platform's impact on adolescent mental health, content moderation practices, and the dissemination of illegal or inappropriate content.

2.3.1 Instagram Map

Despite its numerous functionalities, Instagram introduced in 2022 a map-based content discovery feature[7]. By anchoring content to existing geographic entities such as cities or others POIPoint of Interest (POI), users can explore both nearby or remote content, as illustrated in Figure 2.4.

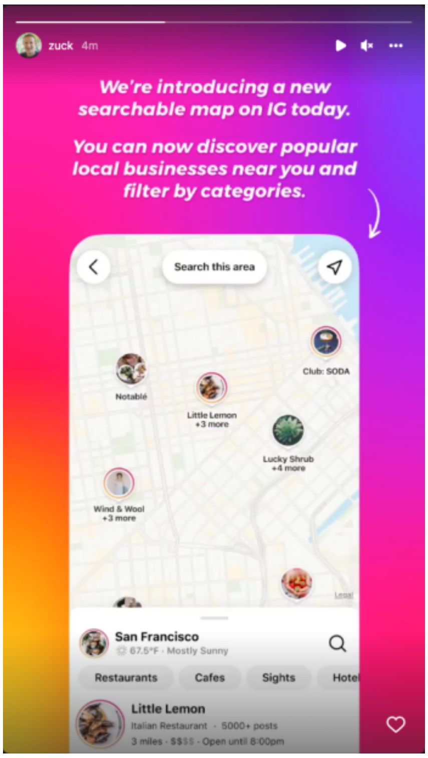

Figure 2.4: Instagram map discovery

In 2026, this functionality underwent a redesign: content is no longer presented as spatial clusters on the map (as implemented in Snapchat, Figure 2.3), instead is represented in a grid-based layout when a specific place is selected via the searchbar of the "Explore" section, as illustrated in Figure 2.5.
In addition, Instagram provides functionality for sharing real-time location with users connections (followers), similarly to other social media platforms discussed in Sections 2.2 and 2.1.

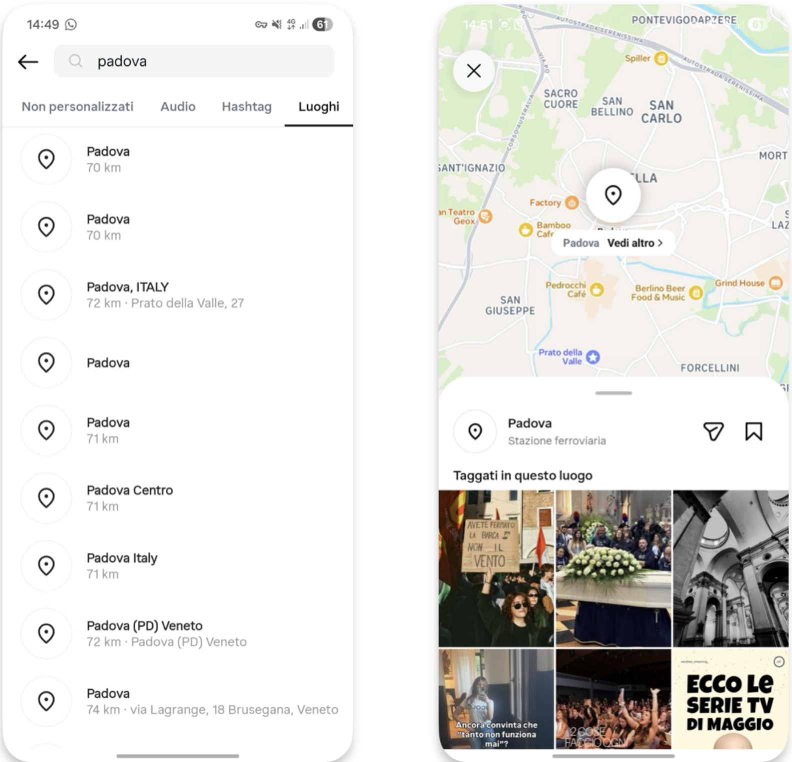

Figure 2.5: Instagram map content 2026

2.4 Google Maps

Google Maps[17] is one of the services provided by Google Inc. Initially released as a web-based platform and following the wide adoption of mobile devices, extended for both iOS and Android systems. The service provides satellite imagery, aerial photography, route planning and 360° interactive panoramic views, commonly known as "Street View"[19]. Google Maps consents users to explore their surroundings by displaying stores, points of interest, and attractions near to their current location or consnet to explorer it scrolling it. The application consents users to explore their surroundings by displaying stores, attractions or others POI POI near their current location, or by enabling them to explore other areas by scrolling the map. These points of interest can be organized into categories such as restaurants, coffee shops, and other predefined types as shown in Figure 2.6. For each place, users may submit reviews by commenting or uploading pictures and videos, thus to create a collaborative information ecosystem. Similarly to Foursquare Swarm 2.1, Google Maps also introduces gamification elements such as badges and user levels to incentivize contributions and improve the quality of place-based reviews.

Google Maps further provides APIs and libraries that allows the development of integrations to its services into thrid-party applications. In fact, the GeoMedia client app 5 through the React Native library adopted, renderized the Google Map as default map framework for Android devices. This service is also well integrated in its ecosystem and Android platforms as well, in fact, exists several standard features, for an instance 'recent places' for places visited by users (within their smartphone) or to have the routing to home or work, enabled and suggested when connected to the car or started in Android Auto [14].

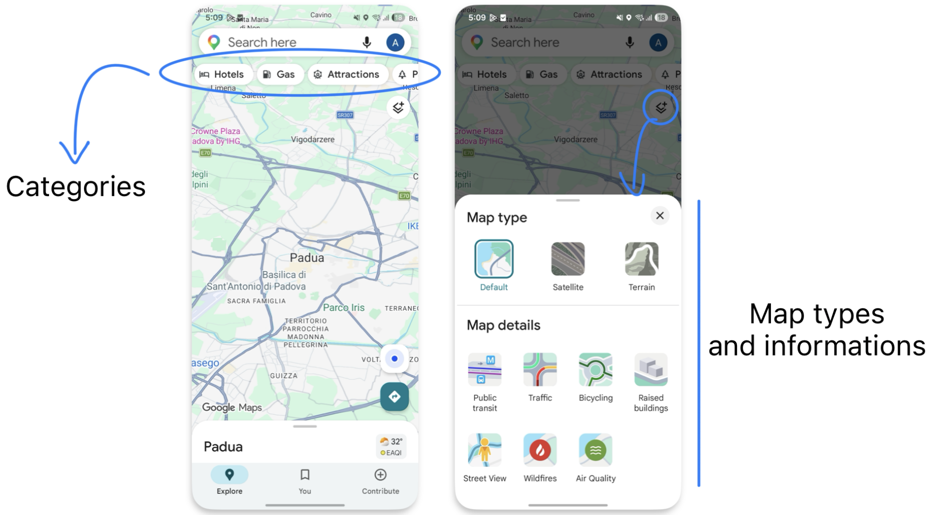

Figure 2.6: Google Maps show case

2.5 Others

There are several other social media platforms that integrate map features and geolocation capabilities. Although they are less popular than the platforms previously discussed, they still cover different scenarios and provide innovative functionalities. For example, Strava, originally developed as an application for tracking running and cycling activities, has evolved into a social networking, by allowing users to share their sports activities, connect with friends, and discover new training partners. While Strava uses geolocation primarily to support sports-related activities, other services adopt location-based interaction at the core of the user experience. More closely related to GeoMedia there are two social media platforms, Jagat [21] and Auglinn [13], both focusing on enabling users to share content and interact with others through a geographical map and current location.

2.5.1 Jagat

Jagat offers several functionalities and highly responsive performance, despite a difficult usability perspective for first-time users, which can appear crowded and confusing, as it displays too many buttons and pieces of information simultaneously, as can be seen in Figure 2.7.
Similar to Snapchat 2.3 or Swarm 2.1, Jagat provide a real-time map enriched with the current location of friends but with more metadata like users' movement speed, or time spent in a specific location or they device battery level. These features raise significant priacy and safety concerns due to the amount of personal information collected and shared.

As social media platform, it also enables users to upload photos and videos, not anchored to any specific location but whenever users want; functionality implemented also by GeoMedia.

Figure 2.7: Jagatt map interface

2.5.2 Auglinn

Available for both Android and iOS platforms, and also for the web, Auglinn is the closest application to GeoMedia. It provides a platform that allows users to create messages enriched with multimedia content and share them with other users through a geographical map. Moreover, Auglinn also offers augmented reality (AR) functionalities to display location-based messages, which can be created either remotely or on demand through a mobile device. It also allows users to create different public, view-only, or private spaces, which act like containers for content (called "notes" or "boxes") and set an expiration time for them, as GeoMedia as well. Other users can respectivaly interact by commenting, saving, liking or sharing notes found.

An example of Auglinn usage is shown in Figure 2.8

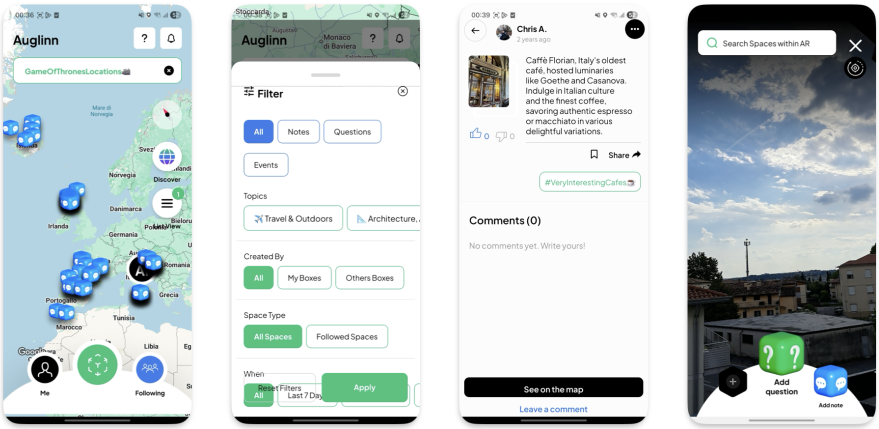

Figure 2.8: Auglinn show case

Chapter 3

GeoMedia

3.1 Idea

3.2 Functionalities

Chapter 4

Architecture and Software design

Chapter 5

Mobile app

Chapter 6

Geomedia Backend

Chapter 7

Conclusions

7.1 Consuntivo finale

7.2 Raggiungimento degli obiettivi

7.3 Conoscenze acquisite

7.4 Valutazione personale

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Padua, May 2026 Alberto Morini

Appendix A

Appendice A

Citazione

Autore della citazione

Acronyms and abbreviations

API Application Program Interface. 20

POI Points of Interest or Point of Interest. 6, 8

Glossary

API in informatica con il termine *Application Programming Interface API* (ing. interfaccia di programmazione di un'applicazione) si indica ogni insieme di procedure disponibili al programmatore, di solito raggruppate a formare un set di strumenti specifici per l'espletamento di un determinato compito all'interno di un certo programma. La finalità è ottenere un'astrazione, di solito tra l'hardware e il programmatore o tra software a basso e quello ad alto livello semplificando così il lavoro di programmazione. 19

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